

AdventureWorks Sales Analytics

SQL Server + Power BI Business Intelligence Project

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1. Executive Summary

This project presents an end-to-end sales analytics solution developed using Microsoft SQL Server and Power BI.

The solution starts with the AdventureWorks transactional database and follows the data through exploration, validation, SQL-based business analysis, transformation, analytical modeling, DAX development, and interactive executive reporting.

The objective was to transform transactional sales data into a structured analytical solution that enables users to monitor sales performance and investigate customers, products, territories, and sales trends.

The final Power BI dashboard provides an executive-level view of the business through key performance indicators, sales trends, product-category analysis, territory performance, and top-selling products.

2. Business Objective

The project was designed to answer key business questions including:

- What is the overall sales performance?
- Which customers contribute the most revenue?
- Which products and categories perform best?
- Which territories generate the highest sales?
- How does sales performance change over time?
- Which products and customers require further investigation?
- How can transactional data be transformed into an interactive decision-support solution?

3. Technology Stack

Technology	Application
SQL Server Express	Database environment
SQL Server Management Studio	Data exploration and SQL analysis
T-SQL	Business and analytical queries
Power Query	Data transformation
Power BI	Data modeling and visualization
DAX	Analytical measures
GitHub	Version control and portfolio documentation

4. Data Analysis

The AdventureWorks database contains transactional information covering sales orders, customers, products, product categories, and sales territories.

Main source tables included:

- Sales.SalesOrderHeader
- Sales.SalesOrderDetail

- Sales.Customer
- Sales.SalesTerritory
- Person.Person
- Production.Product
- Production.ProductSubcategory
- Production.ProductCategory

The initial analysis focused on understanding the database structure, relationships, transaction periods, and appropriate data sources for reporting.

5. SQL Analysis

Eight SQL analysis modules were developed to organize the analytical work.

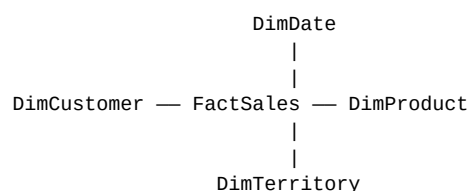
- Data exploration
- Business KPIs
- Product analysis
- Customer analysis
- Territory analysis
- Time-series analysis
- Advanced SQL analysis
- Monthly trend and period-over-period analysis

Advanced SQL techniques included:

- Joins
- Aggregations
- Subqueries
- Common Table Expressions
- CASE
- Window functions
- ROW_NUMBER()
- RANK()
- DENSE_RANK()
- PARTITION BY
- Running totals
- Moving averages
- LAG()

6. Analytical Data Model

The Power BI model uses a fact-and-dimension structure centered around the sales transactions.



The model separates transactional sales data from descriptive dimensions, allowing users to analyze sales consistently across dates, customers, products, and territories.

7. Power Query

Power Query was used to prepare the data for analytical modeling.

- Selecting relevant columns
- Setting appropriate data types
- Creating dimension tables
- Merging customer and person information
- Creating consolidated customer names
- Preparing data for the Power BI model

8. DAX & Business Metrics

DAX measures were developed to provide reusable business metrics within the Power BI model.

- Total Sales
- Total Quantity
- Total Orders
- Average Order Value
- Unique Customers
- Unique Products
- Previous-Year Sales
- Year-over-Year Sales Performance

9. Executive Dashboard

The final Power BI dashboard provides an interactive view of sales performance.

Key Performance Indicators

KPI	Result
Total Sales	\$109.8M
Total Quantity	275K
Total Orders	31,465
Average Order Value	\$3.49K
Unique Customers	19,119
Unique Products	266

Dashboard Components

- Monthly Sales Pattern
- Sales by Category
- Sales by Territory
- Top 10 Selling Products
- Year filter

- Product Category filter
- Product Subcategory filter
- Territory filter

10. Key Insights

Product performance: Bikes represent the dominant product category in the overall sales mix.

Territory performance: Sales vary across territories, providing a basis for comparing geographic market performance.

Product concentration: The Top 10 Selling Products view highlights products making significant contributions to sales.

Sales trends: The monthly sales view makes changes in sales activity visible over time and provides a basis for investigating stronger and weaker periods.

11. Business Value

The solution transforms transactional data into a practical decision-support tool.

- Monitor sales performance
- Compare product and category performance
- Evaluate territory performance
- Identify important customers and products
- Analyze sales trends
- Filter results by key business dimensions

The analytical foundation could be extended into customer segmentation, forecasting, profitability analysis, and more advanced performance optimization.

12. Skills Demonstrated

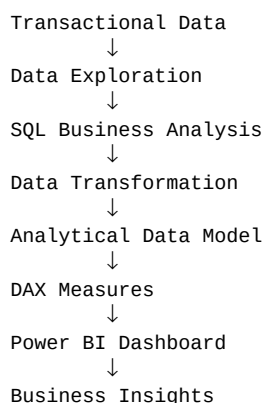
Technical: SQL Server, T-SQL, Power Query, Power BI, DAX, data modeling, data visualization, GitHub.

Analytical: Data exploration, data validation, KPI development, customer analysis, product analysis, territory analysis, time-series analysis, business question formulation.

Business Intelligence: Executive reporting, interactive dashboard development, decision-support analytics, translating transactional data into business insights.

13. Project Outcome

The project demonstrates an end-to-end ability to move from transactional data to business intelligence:



The resulting solution demonstrates how SQL, data modeling, visualization, and business analysis can be combined to support data-driven decision-making.

14. Repository

The complete project repository contains the SQL analysis files, Power BI report, project screenshots, and supporting documentation.